

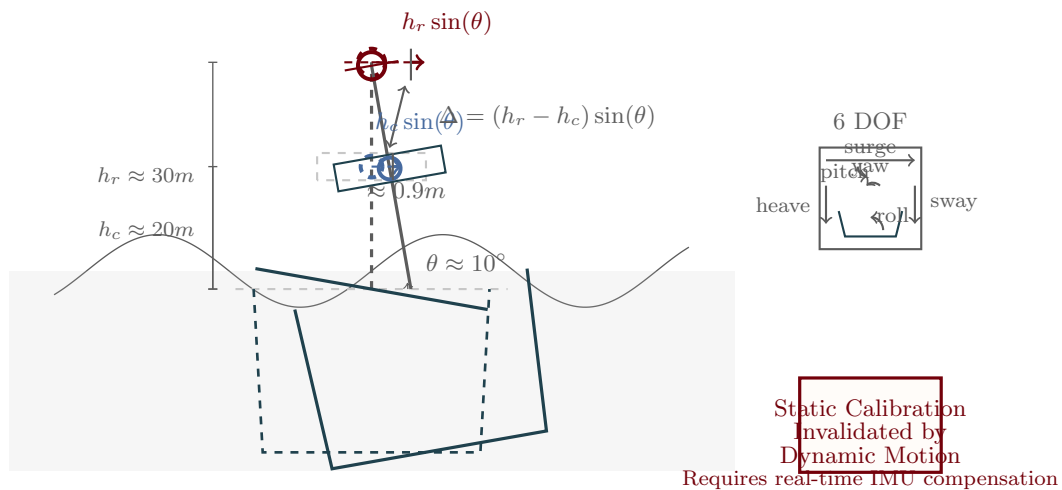


Figure 1: Platform dynamics and lever-arm effects in maritime radar-camera calibration. Vessel roll causes differential lateral displacement between the radar antenna (mounted at mast height h_r) and the camera (at bridge height h_c), introducing time-varying extrinsic calibration errors that must be compensated using inertial measurements.